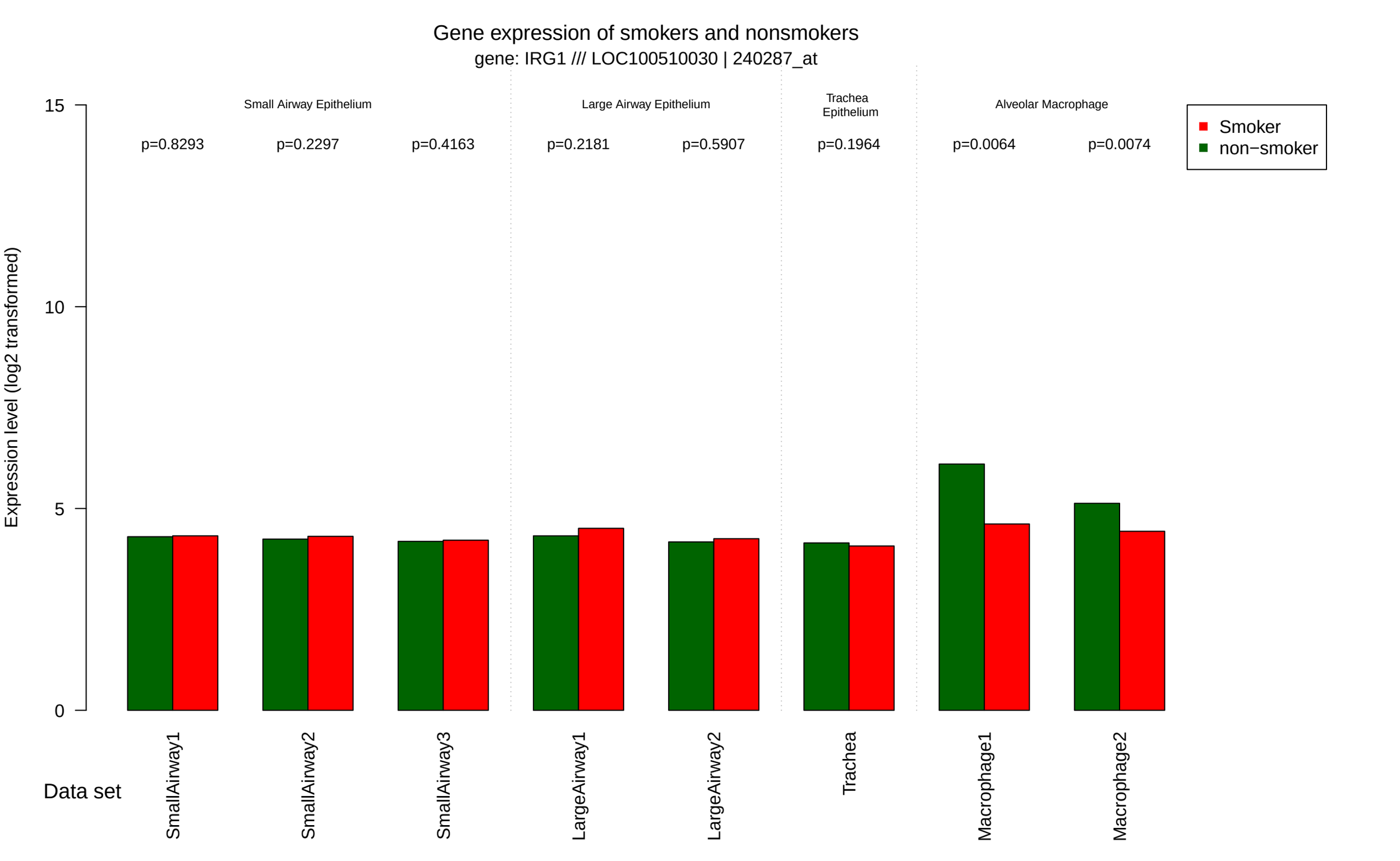
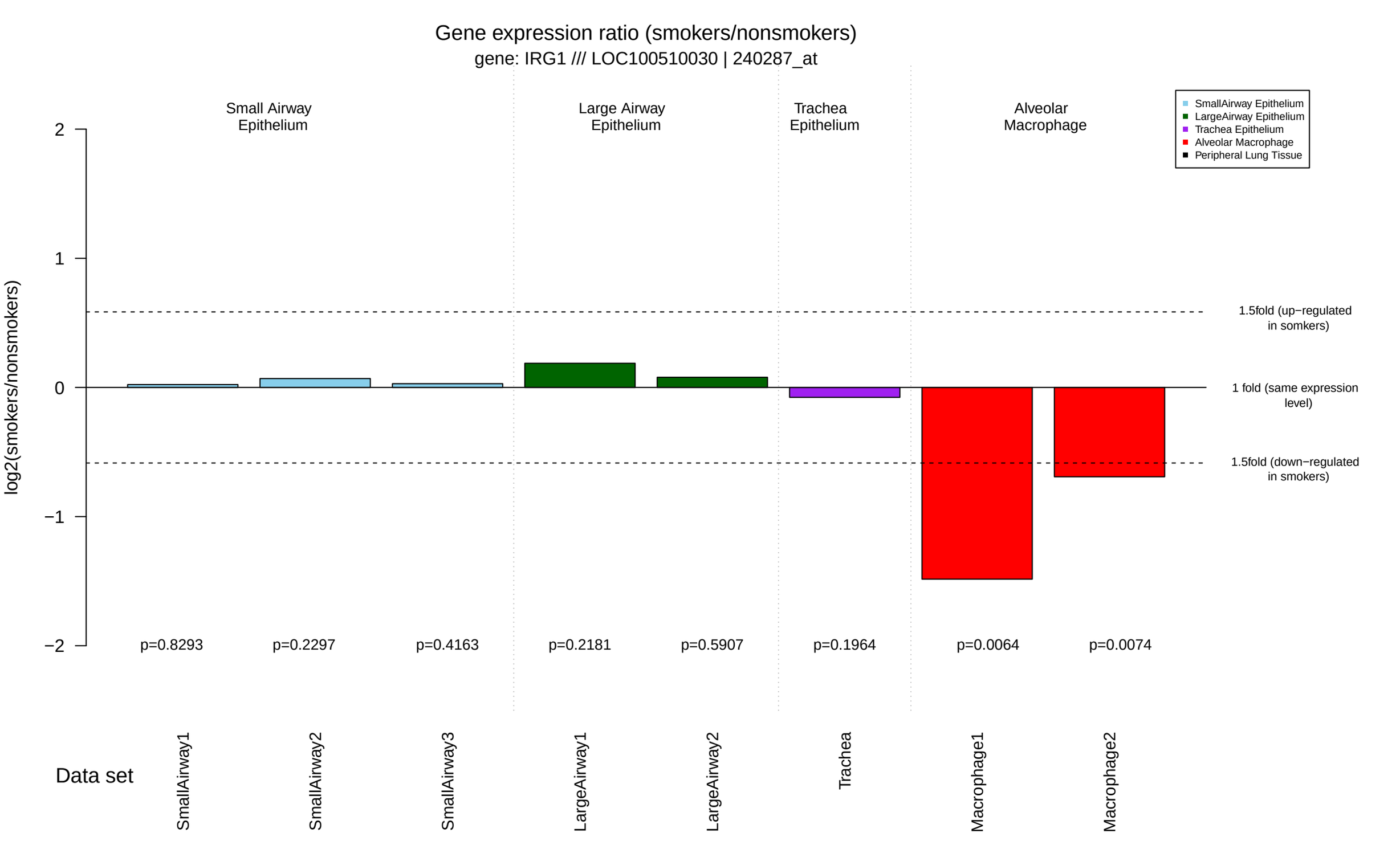
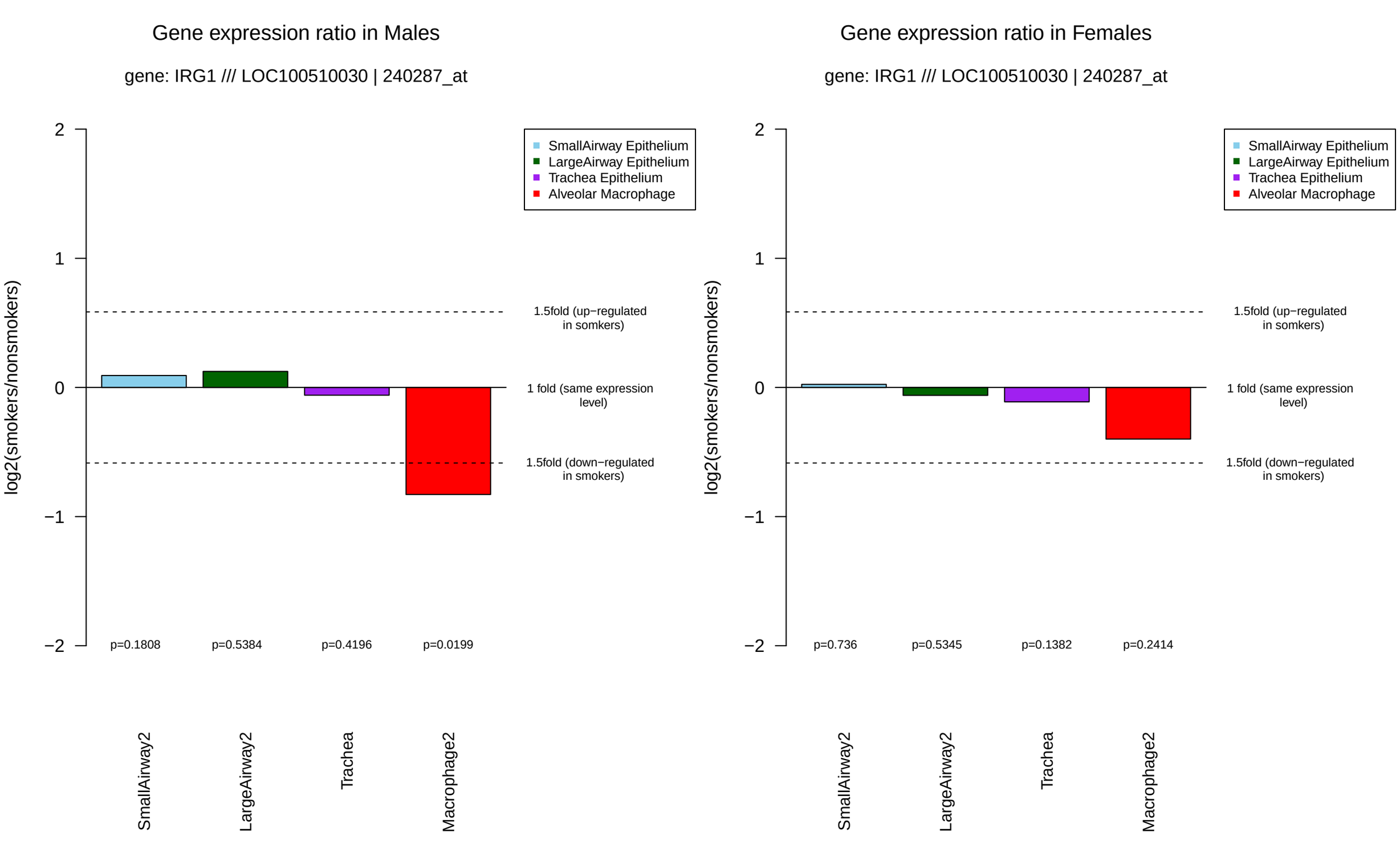
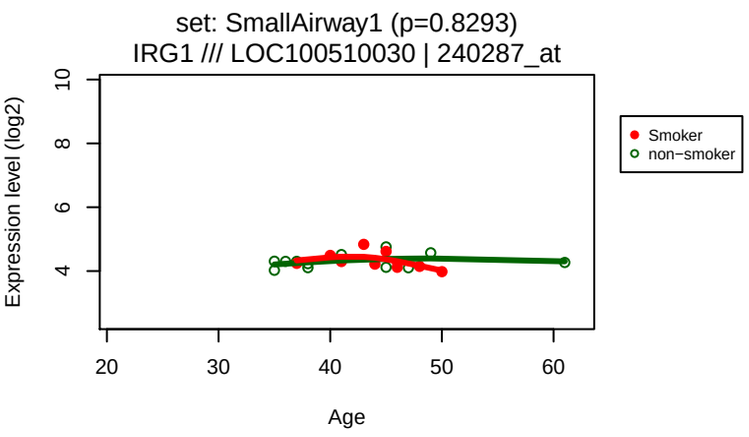
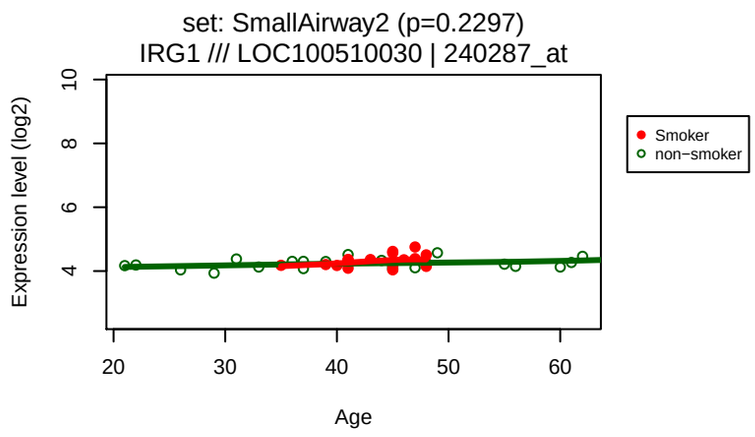
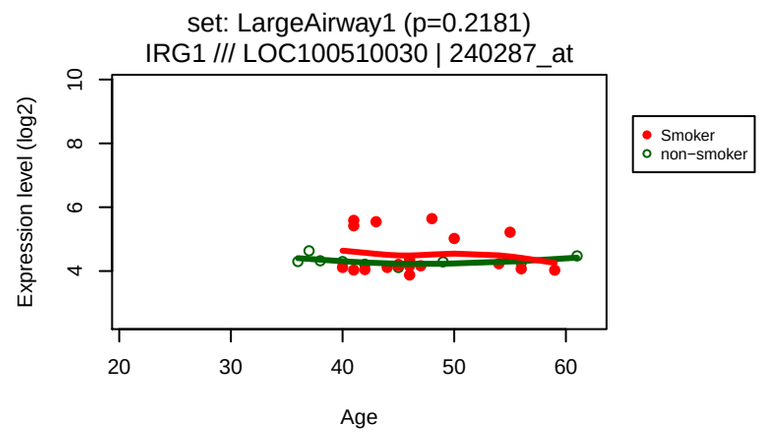

Gene expression of smokers and nonsmokers
gene: IRG1 /// LOC100510030 | 240287_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8293
p=0.2297
p=0.4163
p=0.2181
p=0.5907
p=0.1964
p=0.0064
p=0.0074
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

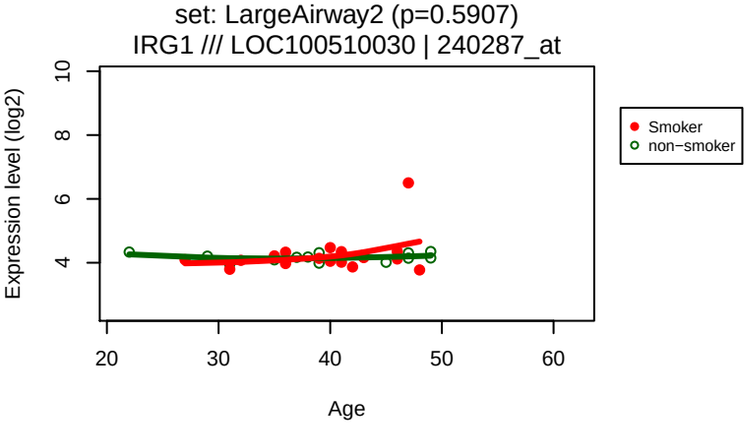

Gene expression ratio (smokers/nonsmokers)
gene: IRG1 /// LOC100510030 | 240287_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8293
p=0.2297
p=0.4163
p=0.2181
p=0.5907
p=0.1964
p=0.0064
p=0.0074
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

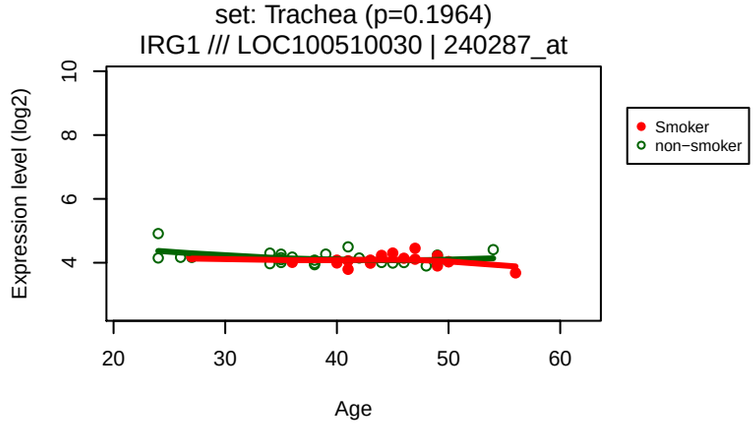

Gene expression ratio in Males
gene: IRG1 /// LOC100510030 | 240287_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1808
p=0.5384
p=0.4196
p=0.0199
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IRG1 /// LOC100510030 | 240287_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.736
p=0.5345
p=0.1382
p=0.2414
SmallAirway2
LargeAirway2
Trachea
Macrophage2

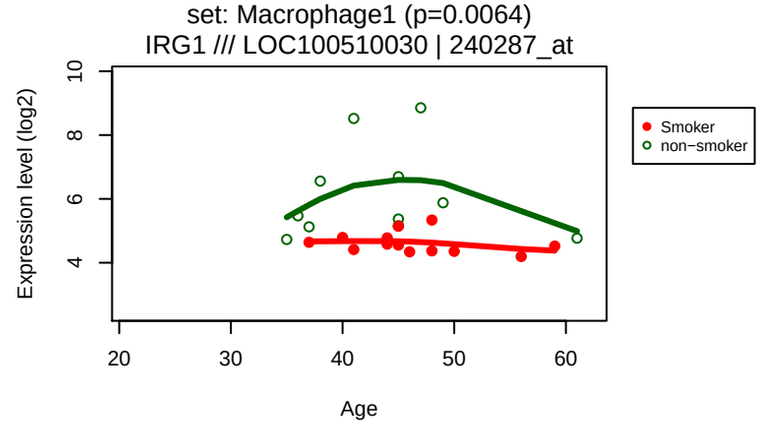

set: SmallAirway1 (p=0.8293)
IRG1 /// LOC100510030 | 240287_at
Expression level (log2)
Age
Smoker
non−smoker

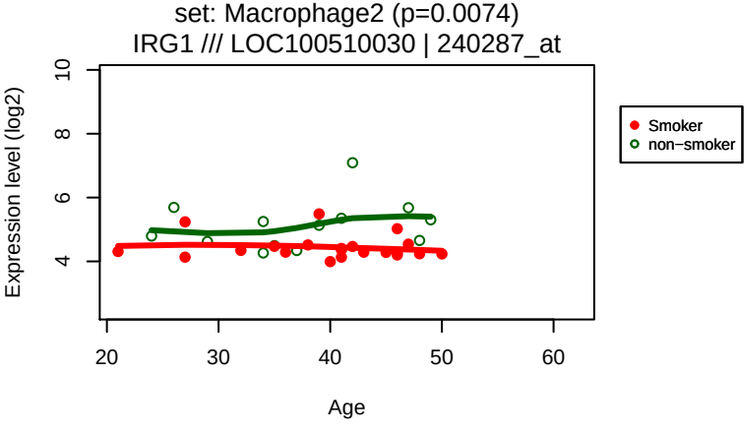

set: SmallAirway2 (p=0.2297)
IRG1 /// LOC100510030 | 240287_at
Expression level (log2)
Age
Smoker
non−smoker

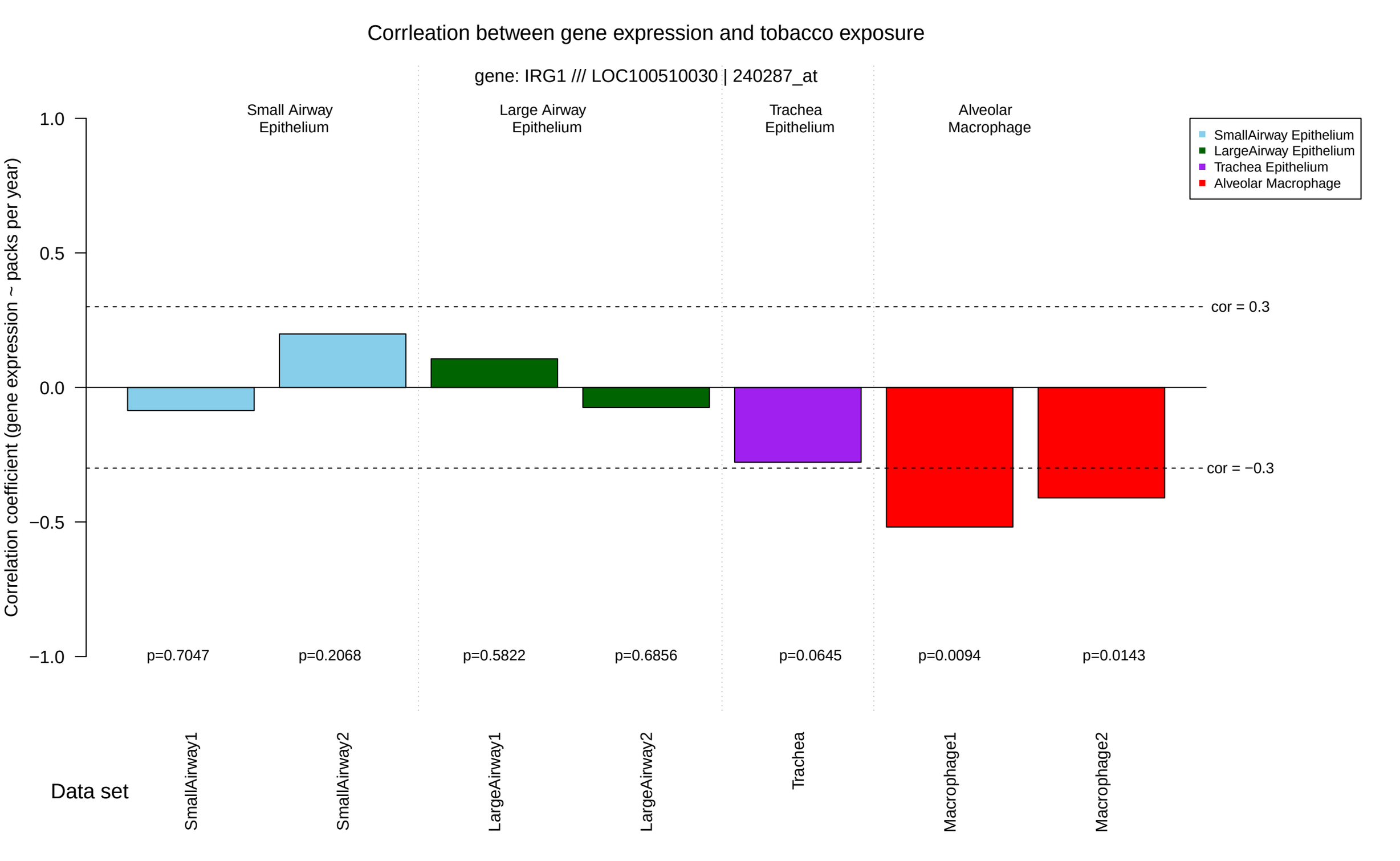

set: LargeAirway1 (p=0.2181)
IRG1 /// LOC100510030 | 240287_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5907)
IRG1 /// LOC100510030 | 240287_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1964)
IRG1 /// LOC100510030 | 240287_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0064)
IRG1 /// LOC100510030 | 240287_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0074)
IRG1 /// LOC100510030 | 240287_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IRG1 /// LOC100510030 | 240287_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7047
p=0.2068
p=0.5822
p=0.6856
p=0.0645
p=0.0094
p=0.0143
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2